



Time: 80 minutes

Full marks: 30

Answer all questions, Show all necessary calculations and interpretations clearly

Marks

1. An engineer collected data on body temperatures (in °C) from five types of mammals, intending to calibrate a biosensor for temperature monitoring. The data is summarized as follows:

Class Interval (°C)	36.5 - 36.9	37.0 - 37.4	37.5 - 37.9	38.0 - 38.4	38.5 - 38.9
Frequency (f)	8	12	10	9	6

- (i) Calculate the median body temperature (°C) of Mammals. 3
(ii) Determine the variance of body temperature (°C) based on the given frequency distribution. 4
(iii) Compute the coefficient of variation (CV). Interpret its meaning in terms of consistency. 3

2. A factory measured the tensile strength (in MPa) of 10 steel rods:
430, 445, 455, 435, 440, 450, 455, 460, 435, 445

- (i) Determine the five-number summary (Minimum, Q1, Median, Q3, Maximum). 4
(ii) Draw a box plot for the tensile strength values. Identify and explain any outliers, if present. 3
(iii) Based on the box plot, comment on the spread and skewness of the tensile strength data. Also write about the advantages of using the box plot. 3

3. An engineering team is analyzing performance metrics from a wireless sensor network deployed in a smart city environment. The following dataset captures the performance of five network nodes in terms of bandwidth and latency. Calculate the strength and direction of the relationship between these two variables (Bandwidth and Latency). 4

Node ID	Bandwidth (Mbps)	Latency (ms)
1	12	50
2	18	45
3	10	60
4	25	30
5	22	35

4. A research team is analyzing the impact of enzyme concentration on reaction rate using a linear regression model. The relationship is described by the following regression equation:
$$2.5 + 0.8x$$
where y = reaction rate ($\mu\text{mol/min}$), and x = enzyme concentration (mg/mL).

- (i) Interpret the slope and intercept of the regression equation in the context of this biochemical process. Predict the reaction rate when enzyme concentration is 5 mg/mL. 3
- (ii) Given observed reaction rates at enzyme concentrations $x=1,2,3$ mg/mL as 5.1, 6.3, 8.05. $\mu\text{mol/min}$ and predicted rates from regression as 5.3, 6.1, 7.9 $\mu\text{mol/min}$. Calculate the residuals (errors) for each data point. 3

Equations

$$r = \frac{N \sum_{i=1}^N X_i Y_i - \sum_{i=1}^N X_i \sum_{i=1}^N Y_i}{\sqrt{\left(N \sum_{i=1}^N X_i^2 - \left(\sum_{i=1}^N X_i \right)^2 \right) \left(N \sum_{i=1}^N Y_i^2 - \left(\sum_{i=1}^N Y_i \right)^2 \right)}}$$

$$b = \frac{n \sum_{i=1}^n X_i Y_i - \sum_{i=1}^n X_i \sum_{i=1}^n Y_i}{n \sum_{i=1}^n X_i^2 - \left(\sum_{i=1}^n X_i \right)^2}$$

$$G = \sqrt[n]{x_1 \cdot x_2 \cdot \dots \cdot x_n}$$

$$G = \text{Antilog} \left(\frac{\sum_{i=1}^n \log X_i}{n} \right)$$

$$H = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}}$$

$$M = L + \left(\frac{\frac{n}{2} - \rho \cdot cf}{f} \right) \cdot h$$

$$M_0 = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \cdot h$$

$$M_0 = L + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (X_i - \mu)^2}{n}$$

$$S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}$$